

LockBench

An Automotive-Radar Chirp-PLL Loop-Filter Designer with
PVT-Robustness Verification and Machine-Learning Inverse Design

Project Report



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Live application: <https://controlbench.onrender.com>

Source: github.com/snehaprasad11/ControlBench

Abstract

Every frequency-modulated continuous-wave (FMCW) automotive radar generates its transmit chirp with a phase-locked loop (PLL); the loop filter that stabilises this PLL is a small passive network whose design fixes the radar's chirp settling time, phase noise and stability, and which must hold specification across the wide automotive temperature range. **LockBench** is a full-stack web tool that, for a *real* commercial synthesizer IC, synthesises this loop filter to target specifications, measures the resulting loop metrics from the closed-loop transfer function, and then verifies that the design still meets specification at every process/voltage/temperature (PVT) corner implied by an AEC-Q100 temperature grade. The synthesizer output is referred up to the 77 GHz radar band through an ideal frequency multiplier, and a random-forest surrogate provides instant inverse design (specifications → design point). The linear PLL model follows Gardner's charge-pump formulation [1]; the PLL-plus-multiplier architecture follows reported 77 GHz FMCW radar transmitters [2].

1. Introduction and Motivation

Automotive radar underpins adaptive cruise control, automatic emergency braking and blind-spot detection. An FMCW radar measures range and velocity by transmitting a linear frequency *chirp* and mixing the echo with the transmitted signal. The chirp is produced by a PLL, and the single component an engineer hand-tunes is the PLL **loop filter** — typically one resistor and two capacitors. Its values set three quantities the radar depends on: how fast the synthesizer settles before each chirp (which limits frame rate and maximum unambiguous velocity), the phase noise (which sets how small a target can be detected next to a large one), and loop stability. Crucially, an automotive part must meet these specifications not at a single laboratory temperature but across the full qualified range (AEC-Q100 grades span $-40\text{ }^{\circ}\text{C}$ to $+150\text{ }^{\circ}\text{C}$), over which the VCO gain and charge-pump current drift substantially. Performing this design and re-verifying it at every temperature corner by hand is slow and error-prone. LockBench automates it and returns a buildable, temperature-qualified design in under a second.

2. Background Theory

2.1 The PLL as a feedback control loop

A charge-pump PLL is a negative-feedback system: a phase-frequency detector and charge pump convert phase error to a current, the loop filter converts that current to the VCO control voltage, the VCO integrates frequency into phase, and a $\pm N$ divider closes the loop. Following Gardner [1], the small-signal open-loop gain is

$$L(s) = [I_{cp} \cdot K_{vco} / (2\pi N)] \cdot Z(s) / s,$$

where I_{cp} is the charge-pump current, K_{vco} the VCO gain (rad/s/V), N the divider ratio and $Z(s)$ the loop-filter transfer impedance. The VCO's $1/s$ and the pole the filter places at the origin make the loop **Type-II** (two poles at $s = 0$), so the loop filter's **zero** is required to recover phase margin — exactly the behaviour Gardner shows is obtainable with a passive filter.

2.2 The loop filter is the design variable

Everything except $Z(s)$ and I_{cp} is fixed by the chosen chip and operating point. LockBench synthesises the passive third-order filter (R , $C1$, $C2$) from two designer targets — loop bandwidth f_c and phase margin ϕ — using the standard closed-form (Banerjee) equations, which place the zero

and high-frequency pole to maximise phase margin at the crossover.

2.3 Measuring rather than assuming

Rather than trusting the closed-form approximations, LockBench builds the resulting transfer functions and *measures* loop bandwidth, phase margin, gain margin, lock (chirp settling) time and jitter peaking numerically. A unit test designs for a target and confirms the measured result matches it, which is what makes the tool trustworthy.

2.4 Phase noise, jitter and frequency multiplication

The output phase noise is modelled as the sum of an in-band contribution (the PLL and reference, anchored to the device datasheet and low-pass shaped by the closed loop) and a VCO contribution (high-pass shaped by the loop error function), integrated over the 12 kHz–20 MHz band into an RMS jitter. The synthesizer output is multiplied to the 77 GHz carrier; ideal $\times N$ multiplication adds $20 \cdot \log_{10}(N)$ to the phase noise while leaving the RMS time jitter invariant — the architecture used in reported 77 GHz radar transmitters [2].

3. Methodology — how a design is produced

1. **Build the loop model** from the selected real device (I_{cp} , K_{vco} , N) and a candidate loop filter.
2. **Synthesise the filter** (R , $C1$, $C2$) from the target loop bandwidth and phase margin.
3. **Measure** phase margin, loop bandwidth, chirp settling and jitter peaking from the transfer functions.
4. **Sweep PVT corners**: hold R/C fixed, perturb K_{vco}/I_{cp} by the drift implied by the chosen AEC-Q100 grade, and re-measure at every corner; flag any spec violation.
5. **Refer to 77 GHz**: apply the multiplier, model phase noise at the RF carrier and integrate the RMS jitter.
6. **Return** buildable R/C values, a pass/fail verdict, per-corner metrics and plot data.

Two additional modes reuse this engine: a **robust-design search** that scans bandwidth/margin/current for a design passing every corner, and an **ML recommend** mode (Section 5).

4. Implementation

LockBench is a single deployed service. A Python engine (built on NumPy and python-control) performs all modelling; a FastAPI backend exposes it as a REST API (`/api/devices`, `/api/design`, `/api/explore`, `/api/recommend`); and a React + Vite frontend renders interactive phase-noise, Bode and lock-transient plots via Plotly.js. The whole application is containerised (Docker) so the backend also serves the built frontend at one URL, deployed on Render. The test suite contains 62 passing tests and runs in continuous integration (GitHub Actions).



Figure 5. The full LockBench web application: inputs and help on the left, live verdict, metrics and plots on the right.

5. Machine-Learning Inverse Design

Design is a forward map (specifications → metrics); engineers want the inverse. A random-forest surrogate is trained on ~900 model-labelled designs sampled across the device library, learning to

predict a design's *worst-case-across-PVT* phase margin, lock time and jitter peaking. Because prediction is near-instant, the inverse-design routine scores a large grid in milliseconds and returns the fastest-settling design that meets the specification. The surrogate is an accelerator over the physics model, not a replacement: the 'Apply' action always re-runs the exact engine to verify the recommendation. On held-out designs the surrogate reproduces worst-case phase margin almost exactly and lock time with modest spread (Figure 4).

6. Results

The figures below are produced directly by the LockBench engine for a 77 GHz long-range-radar profile (TI LMX2594 at 9.625 GHz, multiplied $\times 8$), designed for a 300 kHz loop bandwidth and 55° phase margin and qualified to AEC-Q100 Grade 1.

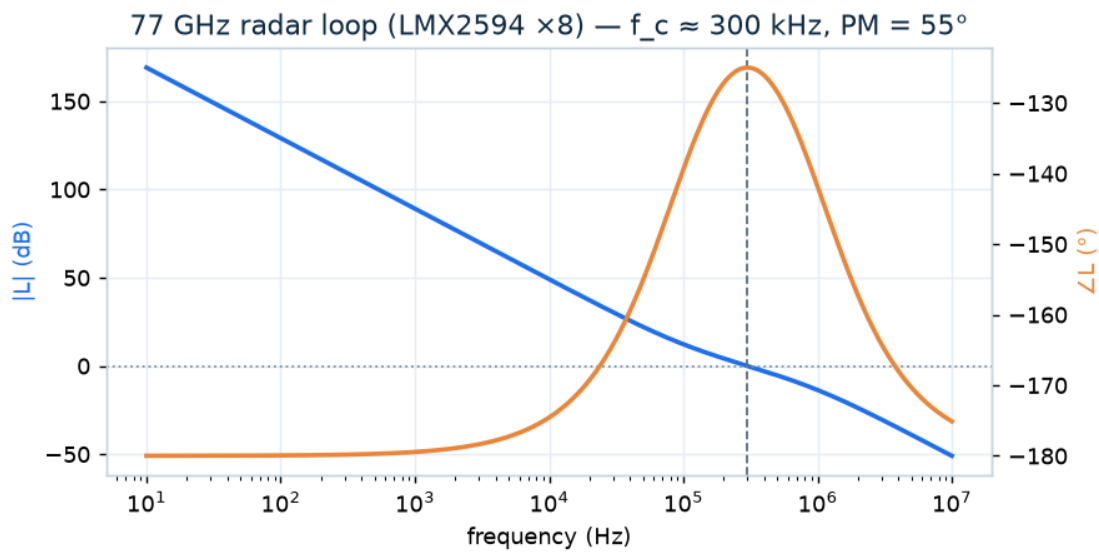


Figure 1. Open-loop Bode response: the loop-filter zero provides the phase bump at crossover that sets the phase margin.

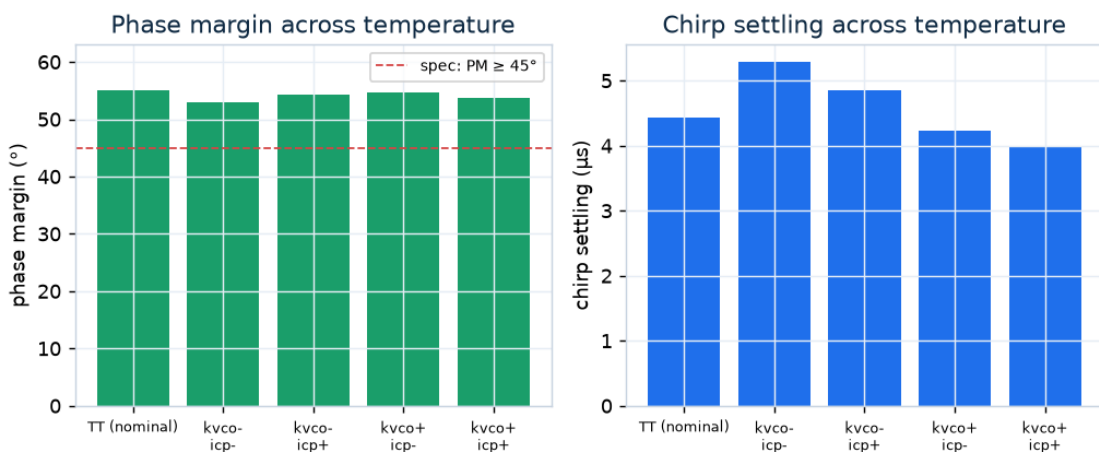


Figure 2. Phase margin and chirp settling across the automotive temperature corners; the worst corner stretches settling but stays within the 45° spec.

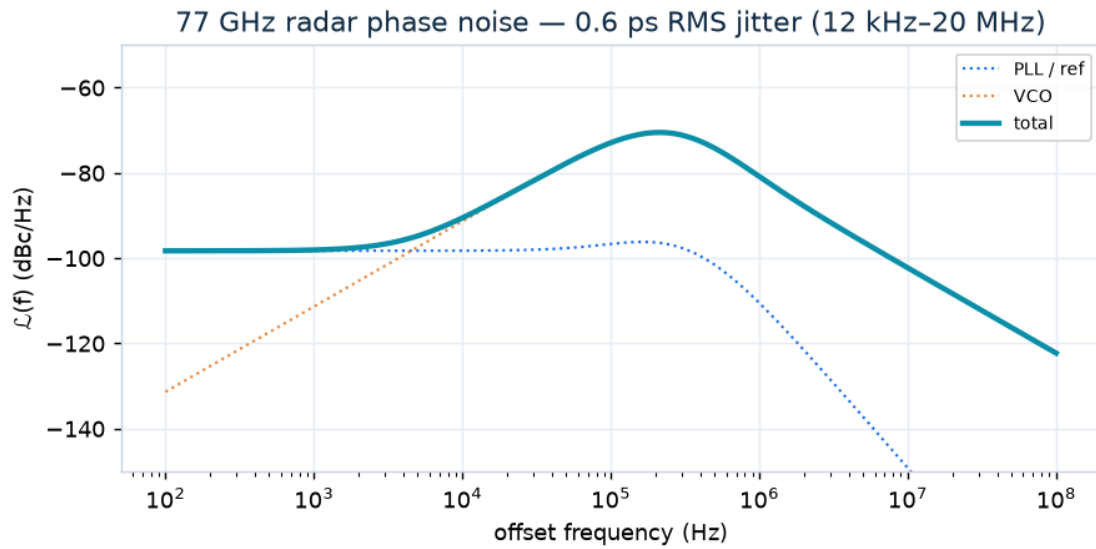


Figure 3. Output phase noise at the 77 GHz carrier (in-band PLL/reference noise, VCO noise, and total) integrated to the RMS jitter.

ML surrogate vs physics model (held-out)

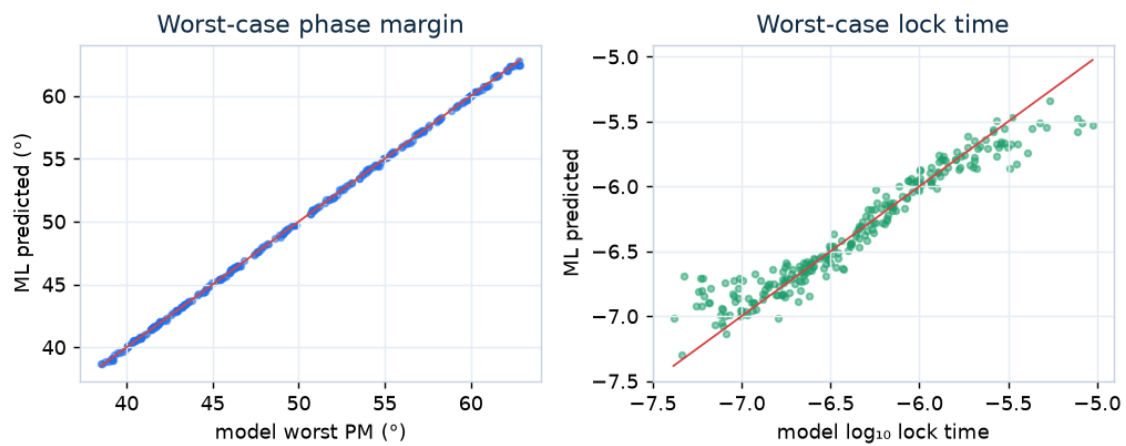
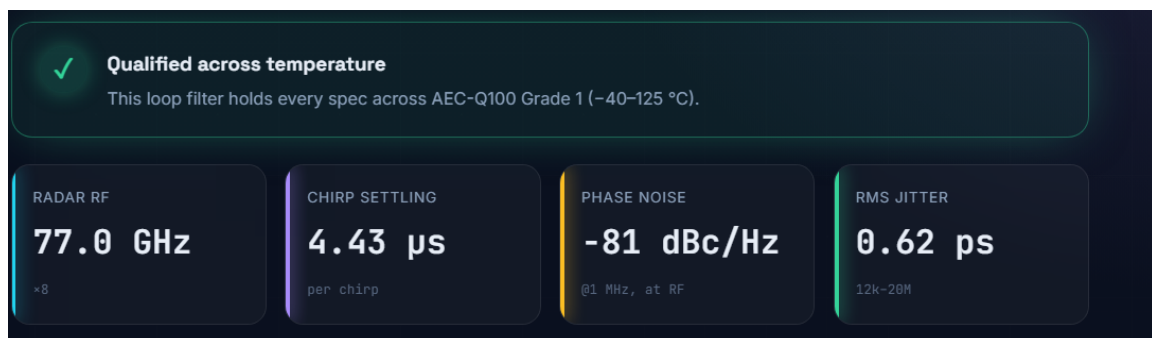
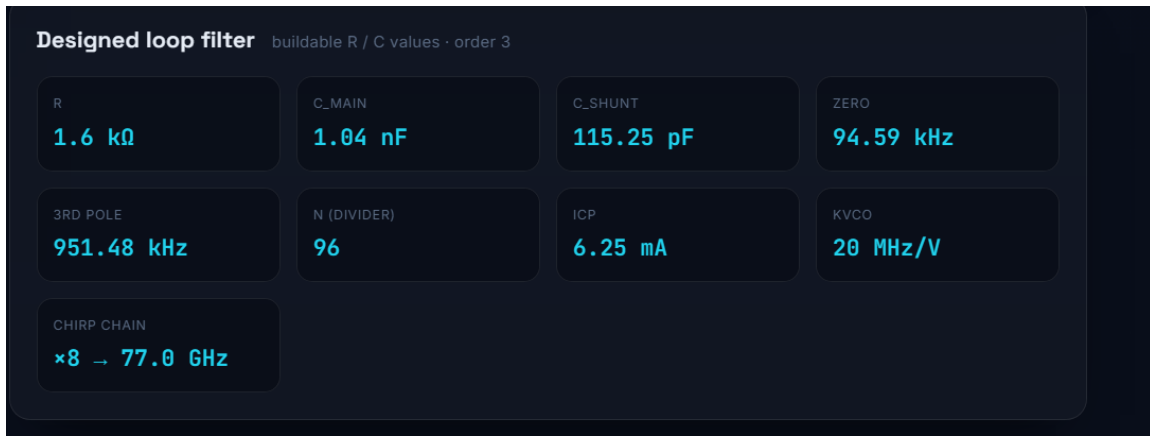


Figure 4. ML surrogate vs. physics model on held-out designs: worst-case phase margin (left) and lock time (right).

The live application presents the same results interactively — a verdict banner, headline metric tiles, the designed component values and the per-corner table:



Verdict banner and headline tiles: radar RF, chirp settling, phase noise and RMS jitter.



The designed loop filter: buildable R / C values, divider, Icp and the $\times 8$ chirp chain to 77 GHz.

PVT corner sweep

CORNER	KVCO	ICP	PM	LOCK	PEAK
TT (nominal)	1.00 $\times$	1.00 $\times$	55.0 $^\circ$	4.43 μ s	2.10
kvco-, icp-	0.70 $\times$	0.90 $\times$	53.0 $^\circ$	5.29 μ s	2.71
kvco-, icp+	0.70 $\times$	1.10 $\times$	54.3 $^\circ$	4.85 μ s	2.42
kvco+, icp-	1.30 $\times$	0.90 $\times$	54.8 $^\circ$	4.23 μ s	1.95
kvco+, icp+	1.30 $\times$	1.10 $\times$	53.8 $^\circ$	3.99 μ s	1.80

The PVT corner sweep: phase margin, lock time and peaking held in-spec at every temperature corner.

7. Novelty and Contributions

- **Anchored on real silicon** — every design uses datasheet parameters of shipping synthesizer ICs (LMX2594, ADF5355/4351), not idealised numbers.
- **Temperature-grade robustness** — unlike a textbook single-point design, LockBench proves the loop holds phase margin, settling and jitter across AEC-Q100 corners; this is the central contribution.
- **Honest ML inverse design** — an accelerator over the verified physics model, always re-checked by the engine.
- **End-to-end, deployed system** — control theory, RF/VLSI modelling, ML and full-stack engineering combined on one safety-critical problem, live at a single URL.

8. Limitations and Scope

The engine uses the continuous-time linear PLL model, valid for loop bandwidths well below the phase-detector frequency; it does not model sampling effects near f_{PFD} or charge-pump non-idealities. Phase noise and jitter are first-order estimates (the VCO term uses a representative profile); the reference-spur figure is an order-of-magnitude estimate; the 77 GHz carrier is reached by an ideal multiplier; and AEC-Q100 grades are mapped to representative $K_{\text{vco}}/I_{\text{cp}}$ drift rather than characterised foundry corners. The ML ground truth is the linear model rather than SPICE or silicon. These are stated plainly in the repository.

9. Conclusion and Future Work

LockBench turns automotive-radar chirp-PLL loop-filter design from a manual, single-temperature calculation into an instant, temperature-qualified, and ML-accelerated workflow grounded in real device data. Natural extensions include band-resolved VCO gain tables, added multiplier noise and spurs, and validating the surrogate against transistor-level (SPICE) simulation rather than the linear model.

References

- [1] F. M. Gardner, “Charge-Pump Phase-Lock Loops,” *IEEE Transactions on Communications*, vol. 28, no. 11, pp. 1849–1858, Nov. 1980.
- [2] “A Low-Phase-Noise 77-GHz FMCW Radar Transmitter With a 12.8-GHz PLL and a $\times 6$ Frequency Multiplier,” *IEEE* (IEEE Xplore document 7488988), 2016. Available: <https://ieeexplore.ieee.org/abstract/document/7488988/>
- [3] B. Liu, “PLL Design for FMCW Radar Systems,” M.S. thesis, EECS, Univ. of California, Berkeley, Tech. Rep. UCB/EECS-2024-106, 2024.
- [4] “A 77-GHz CMOS FMCW Frequency Synthesizer with Reconfigurable Chirps,” *IEEE Trans. Microwave Theory and Techniques*.
- [5] Analog Devices, *ADF4351 / ADF5355 Data Sheets*; Texas Instruments, *LMX2594 Data Sheet*.